Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A and B have different dimensions. Then which of the following relation will be meaningful?

(A) $\left[\frac{A}{B}\right]$

(B) $[A-B]$

(C) $[A+B]$

(D) $\left[e^{A/B}\right]$

Q.2

In order to measure physical quantities in the sub-atomic world, the quantum theory often employs energy [E], angular momentum [J] and velocity [c] as fundamental dimensions instead of the usual mass, length and time. Then, the dimension of pressure in this theory is

(A) $\frac{[\mathrm{E}]^4}{[\mathrm{J}]^3[\mathrm{c}]^3}$

(B) $\frac{[\mathrm{E}]^2}{[\mathrm{J}][\mathrm{c}]}$

(C) $\frac{[\mathrm{E}]}{[\mathrm{J}]^2[\mathrm{c}]^2}$

(D) $\frac{[\mathrm{E}]^3}{[\mathrm{J}]^2[\mathrm{c}]^2}$

Q.3

Of the following quantities, which one has dimensions different from the remaining three?

(A) Energy per unit volume

(B) Force per unit area

(C) Product of voltage and charge per unit volume

(D) Angular momentum

Q.4

The percentage error in measuring M, L and T are 1%, 1.5% and 3% respectively. Then the percentage error in measuring the physical quantity with dimensions $\mathrm{ML}^{-1}\ \mathrm{T}^{-1}$ is

(A) 1%

(B) 3.5%

(C) 3%

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(D) 5.5%

Q.5

If the capacitance of a nanocapacitor is measured in terms of a unit ' u ' made by combining the electric charge ' e ', Bohr radius ' a_0 ', Planck's constant 'h'and speed of light ' c ' then

(A) $\mathrm{u}=\frac{\mathrm{e}^2 \mathrm{~h}}{\mathrm{a}_0}$

(B) $\mathrm{u}=\frac{\mathrm{hc}}{\mathrm{e}^2 \mathrm{a}_0}$

(C) $u=\frac{e^2 c}{h a_0}$

(D) $\mathrm{u}=\frac{\mathrm{e}^2 \mathrm{a}_0}{\mathrm{hc}}$

Q.6

The dimensions of coefficient of self inductance
are

(A) $\left[\mathrm{ML}^2 \mathrm{~T}^{-2} \mathrm{~A}^{-2}\right]$

(B) $\left[\mathrm{ML}^2 \mathrm{~T}^{-2} \mathrm{~A}^{-1}\right]$

(C) $\left[\mathrm{MLT}^{-2} \mathrm{~A}^{-2}\right]$

(D) $\left[\mathrm{MLT}^{-2} \mathrm{~A}^{-1}\right]$

Q.7

A quantity X is given by $\varepsilon_0 L \frac{\Delta V}{\Delta t}$ where ϵ_0 is the permittivity of the free space, L is a length, ΔV is a potential difference and Δt is a time interval. The dimensional formula for X is the same as that of

(A) resistance

(B) charge

(C) voltage

(D) current

Q.8

If Q denote the charge on the plate of a capacitor of capacitance C then the dimensional formula for $\frac{\mathrm{Q}^2}{\mathrm{C}}$ is

(A) $\left[\mathrm{L}^2 \mathrm{M}^2 \mathrm{~T}\right]$

(B) $\left[\mathrm{LMT}^2\right]$

(C) $\left[\mathrm{L}^2 \mathrm{MT}^{-2}\right]$

(D) $\left[\mathrm{L}^2 \mathrm{M}^2 \mathrm{~T}^2\right]$

Q.9

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Which one of the following represents the correct dimensions of the coefficient of viscosity?

(A) $\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-1}\right]$

(B) $\left[\mathrm{MLT}^{-1}\right]$

(C) $\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-2}\right]$

(D) $\left[\mathrm{ML}^{-2}\ \mathrm{T}^{-2}\right]$

Q.10

The physical quantities not having same dimensions are

(A) torque and work

(B) momentum and Planck's constant

(C) stress and Young's modulus

(D) speed and $\left(\mu_0 \varepsilon_0\right)^{-1/2}$

Q.11

An object is moving through the liquid. The viscous damping force acting on it is proportional to the velocity. Then dimensions of constant of proportionality are

(A) $\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-1}\right]$

(B) $\left[\mathrm{MLT}^{-1}\right]$

(C) $\left[\mathrm{M}^{0}\mathrm{LT}^{-1}\right]$

(D) $\left[\mathrm{ML}^{0}\ \mathrm{T}^{-1}\right]$

Q.12

A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{\mathrm{e}^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge]

(A) $\mathrm{c}^2\left[\mathrm{G}\frac{\mathrm{e}^2}{4\pi\varepsilon_0}\right]^{1/2}$

(B) $\frac{1}{\mathrm{c}^2}\left[\frac{\mathrm{e}^2}{\mathrm{G}4\pi\varepsilon_0}\right]^{1/2}$

(C) $\frac{1}{c}\mathrm{G}\frac{\mathrm{e}^2}{4\pi\varepsilon_0}$

(D) $\frac{1}{\mathrm{c}^2}\left[\mathrm{G}\frac{\mathrm{e}^2}{4\pi\varepsilon_0}\right]^{1/2}$

Q.13

The resistance of a metal is given by $R = \frac{V}{I}$, where V is potential difference and I is the current. In a circuit the potential difference across resistance is $V = (8 \pm 0.5)\mathrm{V}$ and current n resistance, $I = (4 \pm 0.2)\mathrm{A}$. What is the value of resistance with its percentage error?

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(A) $(2 \pm 5.6\%)\Omega$

(B) $(2 \pm 0.7\%)\Omega$

(C) $(2 \pm 35\%)\Omega$

(D) $(2 \pm 11.25\%)\Omega$

Q.14

A highly rigid cubical block A of small mass M and side L is fixed rigidy on the othe-cubical block of same dimensions and of low modulus of rigidity η such that the lower face of A completely covers the upper face of B. The lower face of B is rigidly held on a horizontal surface. A small force F is applied perpendicular to one of the side faces of A. After the force is withdrawn, block A executes small oscillations, the time period of which is given by

(A) $2\pi\sqrt{M\pi L}$

(B) $2\pi\sqrt{(Mn/L)}$

(C) $2\pi\sqrt{ML/n}$

(D) $2\pi\sqrt{M/\eta L}$

Q.15

A spherical body of mass m and radius r is allowed to fall in a medium of viscosity η. The time in which the velocity of the body increases from zero to 0.63 times the terminal velocity (v) is called time constant (\tau). Dimensionally, τ can be represented by

(A) $\frac{m^2}{6\pi n}$

(B) $\sqrt{\left(\frac{6mm\eta}{g^2}\right)}$

(C) $\frac{m}{6\pi nm}$

(D) None of these

Q.16

Which of the following product of e, h, μ, G (where μ is the permeability) be taken so that the dimensions of the product are same as that of speed of light ?

(A) $he^{-2}\mu^{-1}G^0$

(B) $h^2eG^{-1}\mu$

(C) $h^0e^2G^{-1}\mu$

(D) $hGe^{-2}\mu^0$

Q.17

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The position x of a particle at time t is given by $x = \frac{V_0}{a}\left(1 - e^{-\alpha t}\right)$, where V_0 is constant and $a > 0$. The dimensions of V_0 and a are

(A) $M^{\theta}LT^{-1}$ and T^{-1}

(B) $M^{0}LT^{\circ}$ and T^{-1}

(C) $M^{0}LT^{-1}$ and LT^{-2}

(D) $M^{0}LT^{-1}$ and T

Q.18

The dimensions of $\frac{1}{2}\varepsilon_0 E^2$ ($\varepsilon_0 =$ permittivity of free space and $E =$ electric field) are

(A) $\left[ML^2T^{-1}\right]$

(B) $\left[ML^{-1}T^{-2}\right]$

(C) $\left[ML^2T^{-2}\right]$

(D) $MLT^{-1}\Big]$

Q.19

A physical quantity X is represented by $\mathrm{X} = \left(\mathrm{M}^{\mathrm{x}}\mathrm{L}^{-y}\ \mathrm{T}^{-z}\right)$. The maximum percentage errors in the measurement of M, L and T, respectively, are $a\%, b\%$ and c%. The maximum percentage error in the measurement of X will be:

(A) $(ax + by - cz)\%$

(B) $(ax - by - cz)\%$

(C) $(ax + by + cz)\%$

(D) $(a - by + cz)\%$

Q.20

Given that: $y = A\sin\left[\left(\frac{2\pi}{\lambda}\right)(ct - x)\right]$, where y and x are measured in the unit of length.
Which of the following statements is true ?

(A) The unit of λ is same as that of x and A

(B) The unit of λ is same as that of x but not of A

(C) The unit of c is same as that of $2\pi/\lambda\mathrm{c}$

(D) The unit of $(ct - x)$ is same as that of $2\pi/\lambda$

Answers & Solutions

Q.1 Answer:

$\left[\frac{A}{B}\right]$

Solution:

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A and B have different dimensions. Hence, only A and B in a ratio form and is meaningful.

Q.2 Answer:

$\frac{[\mathrm{E}]^4}{[\mathrm{J}]^3[\mathrm{c}]^3}$

Solution:

$[\mathrm{E}] = \left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right]$

$[\mathrm{J}] = \left[\mathrm{ML}^2\ \mathrm{T}^{-1}\right]$

(ii) . . . . . (i

$[\mathrm{C}] = \left[\mathrm{LT}^{-1}\right]$

(iii) Solving (i), (ii) and (iii) we get,

$\left[\frac{\mathrm{E}}{\mathrm{C}^2}\right] = [\mathrm{M}], \quad \left[\frac{\mathrm{JC}}{\mathrm{E}}\right] = [\mathrm{L}]$ **and** $\left[\frac{\mathrm{J}}{\mathrm{E}}\right] = [\mathrm{T}]$

Now, [Pressure] $= \left[\mathrm{ML}^{-1}\ \mathrm{T}^{-2}\right]$

$= \left[\frac{\mathrm{E}}{\mathrm{C}^2}\right] \times \left[\frac{\mathrm{E}}{\mathrm{JC}}\right] \times \left[\frac{\mathrm{E}^2}{\mathrm{J}^2}\right] = \frac{[\mathrm{E}]^4}{[\mathrm{J}^3][\mathrm{C}^3]}$

Q.3 Answer:

Angular momentum

Solution:

For angular momentum, the dimensional formula is $\left[\mathrm{ML}^2\ \mathrm{T}^{-1}\right]$**. For other three, it is** $\left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right]$

Q.4 Answer:

5.5%

Solution:

Let $\mathrm{X} = \left[\mathrm{ML}^{-1}\ \mathrm{T}^{-1}\right]$

Then, $\frac{\Delta \mathrm{X}}{\mathrm{X}} \times 100 = \left(\frac{\Delta \mathrm{M}}{\mathrm{M}} + \frac{\Delta \mathrm{L}}{\mathrm{L}} + \frac{\Delta \mathrm{T}}{\mathrm{T}}\right) \times 100$

As we know, $\frac{\Delta \mathrm{M}}{\mathrm{M}} = 1\%$, $\frac{\Delta \mathrm{L}}{\mathrm{L}} = 1.5\%$ **and** $\frac{\Delta \mathrm{T}}{\mathrm{T}} = 3\%$

$= (1 + 1.5 + 3)\% = 5.5\%$

Q.5 Answer:

$\mathrm{u} = \frac{\mathrm{e}^2 \mathrm{a}_0}{\mathrm{hc}}$

Solution:

Let unit ' u **' related with** e, a_0, h **and** c **as follows.** $[u] = [e]^a\,[a_0]^b\,[h]^c[C]^d$

Using dimensional method, $\left[M^{-1}L^{-2}T^{+4}A^{+2}\right] = \left[A^1T^1\right]^a\,[L]^b\,\left[ML2T^{-1}\right]^c\,\left[LT^{-1}\right]^d$

$\left[M^{-1}L^{-2}T^{+4}A^{+2}\right] = \left[M^cL^{b+2c+d}T^{a-c-d}A^a\right]$

$a = 2, b = 1, c = -1, d = -1$

$\therefore \quad u = \frac{e^2 a_0}{hc}$

Q.6 Answer:

$\left[\mathrm{ML}^2\ \mathrm{T}^{-2}\ \mathrm{A}^{-2}\right]$

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Solution:

Energy stored in an inductor, $U = \frac{1}{2}LI^2 \Rightarrow \mathrm{L} = \frac{2\mathrm{U}}{\mathrm{I}^2}$

$[\mathrm{L}] = \frac{\left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right]}{[\mathrm{A}]^2} = \left[\mathrm{ML}^2\ \mathrm{T}^{-2}\ \mathrm{A}^{-2}\right]$

Q.7 Answer:

current

Solution:

Dimensionally $\varepsilon_0 L =$ **Capacitance (c)**

$\therefore \varepsilon_0 L\frac{\Delta V}{\Delta t} = \frac{C\Delta V}{\Delta t} = \frac{q}{\Delta t} = I$

Q.8 Answer:

$\left[\mathrm{L}^2\mathrm{MT}^{-2}\right]$

Solution:

We know that $\frac{\mathrm{Q}^2}{2\mathrm{C}}$ **is energy of capacitor so it represent the dimension of energy** $= \left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right]$

Q.9 Answer:

$\left[\mathrm{ML}^{-1}\ \mathrm{T}^{-1}\right]$

Solution:

From stokes law $\mathrm{F} = 6\pi\eta\mathrm{rv} \Rightarrow \eta = \frac{\mathrm{F}}{6\pi\mathrm{rv}}$

$\therefore \eta = \frac{\mathrm{MLT}^{-2}}{[\mathrm{L}][\mathrm{LT}^{-1}]} \Rightarrow \eta = \left[\mathrm{ML}^{-1}\ \mathrm{T}^{-1}\right.$.

Q.10 Answer:

momentum and Planck's constant

Solution:

$P = [\mathrm{M}][\mathrm{L}]\left[\mathrm{T}^{-1}\right] = \left[\mathrm{MLT}^{-1}\right]$

$\mathrm{P} =$ **momentum**

Planck's constant $= \frac{\mathrm{E}}{\mathrm{v}} = \frac{[\mathrm{M}]\left[\mathrm{LT}^{-1}\right]^2}{\mathrm{T}^{-1}} = \left[\mathrm{ML}^2\ \mathrm{T}^{-1}\right]$

Option (b) is correct.

Q.11 Answer:

$\left[\mathrm{ML}^0\ \mathrm{T}^{-1}\right]$

Solution:

$F \propto v \Rightarrow F = kv \Rightarrow [k] = \left[\frac{F}{v}\right] = \left[\frac{MLT^{-2}}{LT^{-1}}\right] = \left[ML^0T^{-1}\right]$

Q.12 Answer:

$\frac{1}{\mathrm{c}^2}\left[\mathrm{G}\frac{\mathrm{e}^2}{4\pi\varepsilon_0}\right]^{1/2}$

Solution:

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Let dimensions of length is related as,

$L = [c]^x [G]^y \left[\frac{e^2}{4\pi\varepsilon_0}\right]^z$

$\frac{\mathrm{e}^2}{4\pi\varepsilon_0} = \mathrm{ML}^3\,\mathrm{T}^{-2}$

$\mathrm{L} = \left[\mathrm{LT}^{-1}\right]^{\mathrm{x}} \left[\mathrm{M}^{-1}\,\mathrm{L}^3\,\mathrm{T}^{-2}\right]^{\mathrm{y}} \left[\mathrm{ML}^3\,\mathrm{T}^{-2}\right]^2$

$[L] = \left[L^{x+3y+3z} M^{-y+z} T^{-x-2y-2z}\right]$

Comparing both sides

$-y + z = 0 \Rightarrow y = z$**.....(i)**

$x + 3y + 3z = 1$**.....(ii)**

$-x - 4z = 0 \quad (\because y = z)$**.....(iii)**

From (i), (ii) and (iii) $z = y = \frac{1}{2}, x = -2$

Hence, $L = c^{-2}\left[G \cdot \frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

Q.13 Answer:

$(2 \pm 11.25\%)\Omega$

Solution:

$R = \frac{V}{l} = \frac{8}{4} = 2\mathrm{ohm}$

$\frac{\Delta \mathrm{R}}{\mathrm{R}} \times 100 = \frac{\Delta \mathrm{V}}{\mathrm{V}} \times 100 + \frac{\Delta \mathrm{I}}{\mathrm{I}} \times 100$

$= \frac{0.5}{8} \times 100 + \frac{0.2}{4} \times 100 = 11.25\%$

$\Rightarrow R = (2 \pm 11.25)\Omega$

Q.14 Answer:

$2\pi\sqrt{M/\eta L}$

Solution:

Correct option is(d) $2\pi\sqrt{\mathrm{M}/\eta\mathrm{L}}$

Modulus of rigidity $\eta = \frac{\mathrm{F}}{\mathrm{A}\theta}$

Here, $\mathrm{A} = \mathrm{L}^2$

and $\theta = \frac{\mathrm{x}}{\mathrm{L}}$ **for small** θ

$\therefore$ **Restoring force** $= \mathrm{F} = -\eta \mathrm{A}\theta$

or acceleration, $a = \frac{\mathrm{F}}{\mathrm{m}} = \frac{\eta \mathrm{L}}{\mathrm{M}}\mathrm{x}$ **equation** (1)

$\because \mathrm{a} \propto (-\mathrm{x})$

$\therefore$ **Time period,** $\mathrm{T} = 2\pi\sqrt{\left|\frac{\mathrm{x}}{\mathrm{a}}\right|}$

$\Longrightarrow \mathrm{T} = 2\pi\sqrt{\frac{\mathrm{M}}{\eta \mathrm{L}}}$

Q.15 Answer:

None of these

Solution:

$\left[\frac{mr^2}{6\pi\eta}\right] = \left[\frac{ML^2}{ML^{-1}T^{-1}}\right] = \left[L^3 T\right]$

As we have $[\eta] = \left[ML^{-1}T^{-1}\right]$

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$\left[\left(\frac{6\pi wr\eta}{g^2}\right)^{\frac{1}{2}}\right]=\left[\left(\frac{MLML^{-1}T^{-1}}{L^2T^{-4}}\right)^{\frac{1}{2}}\right]$

$\left[\frac{m}{6\pi\eta rv}\right]=\left[\frac{M}{ML^{-1}T^{-1}LLT^{-1}}\right]=\left[L^{-1}T^{-2}\right]$

Thus, none of the given expressions have the dimensions of time.

Q.16 Answer:

$he^{-2}\mu^{-1}G^0$

Solution:

Here $v=e^ah^b\mu^cG^d$. **Taking the dimensions**

$M^0LT^{-1}A^0=[AT]^a\left[ML^2T^{-1}\right]^b$

$\left[MLT^{-2}A^2\right]^\rho\left[M^{-1}LT^{-2}\right]^d$

There will be 4 simultaneous equations by equating the dimensions of M, L, T **and** A. **These are** $a-2c=0, a-b-2c-2d=-1, b+c-d=0$ **and** $2b+c+3d=1$

Solving for a, b, c and d, we get

$a=-2, b=1, c=-1, d=0$

Thus, $v=e^{-2}hu^{-1}G^0$

Q.17 Answer:

$M^\theta LT^{-1}$ and T^{-1}

Solution:

(a)

Here, at **is dimensionless**

$\Rightarrow a=\frac{1}{t}=\left[\frac{1}{T}\right]=\left[T^{-1}\right]$

$x=\frac{V_0}{a}$ **and** $V_0=xa=\left[LT^{-1}\right]=\left[M^0LT^{-1}\right]$

Q.18 Answer:

$\left[ML^{-1}T^{-2}\right]$

Solution:

Explanation:

Here $(1/2)\varepsilon_0\mathrm{E}^2$ **represents energy per unit volume**

$[\varepsilon_0]\left[\mathrm{E}^2\right]=$ **Energy / volume** $=\left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right]/\left[\mathrm{L}^3\right]$

$=\mathrm{ML}^{-1}\ \mathrm{T}^{-2}$

Q.19 Answer:

$(ax+by+cz)\%$

Solution:

$X=M^xL^{-y}T^{-z}$

$\therefore\frac{\Delta X}{X}\times100=x\left(\frac{\Delta M}{M}\times100\right)+y\left(\frac{\Delta L}{L}\times100\right)+z\left(\frac{\Delta T}{T}\times100\right)$ **(Errors are always added)**

$\therefore\frac{\Delta X}{X}\times100=(\alpha x+by+cz)$ **per cent.**

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Q.20 Answer:

The unit of λ is same as that of x and A

Solution:

Unit of ct must be same as unit of x. Since argument of sin has to be dimensionless, thus dimensions of x and ct must be same as dimensions of λ. Thus dimension of λ equals 'L'. Also, dimension of y equals 'L' and hence dimension of A equals 'L' because sin term is dimensionless. Thus statement 1 is correct.

All others are incorrect:

Dimension of ct-x is same as λ, not $\frac{1}{\lambda}$. Thus D is incorrect. C is incorrect because dimension of c is same as dimension of $\frac{\lambda}{t}$.

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